

# Youssef Sghairi

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LinkedIn | Github | Portfolio

## Profile

Final-year Software Engineering student at INSAT, seeking an end-of-studies internship. Full-stack experience shipping production features across React/Next.js frontends, Python/Java/Go backends, and event-driven architectures, with hands-on work on AI-assisted workflows and multi-tenant systems. Comfortable owning a feature from technical design through implementation, testing, and release.

## Experience

### SpecTraceAI – Full Stack Engineer UAE

Mar 2026 – Present  
Remote

- Designed a provider-agnostic ALM integration framework and shipped its first implementation for Polarion, enabling secure bidirectional sync of requirements, test cases, and LiveDoc views, with extensibility for future ALM providers.
- Contributed to an AI-assisted repository discovery workflow that explores monorepos, identifies engineering artifacts, supports human review, and persists approved mappings as project configuration.
- Designed deterministic test doubles for LLM, embeddings, and vector-store integrations, improving repeatability and reducing reliance on external Azure OpenAI and Pinecone calls.
- Designed and implemented multi-tenant project support with role-based access control, enabling multiple users to securely collaborate on shared projects.

### Uvey – Software Developer Tunis, Tunisia

Jun 2025 – Mar 2026  
Hybrid

- Delivered end-to-end product features using React, Quarkus, and Kafka translating evolving client requirements into maintainable production software.
- Migrated a service from an existing microservice into a standalone service, transitioning its persistence layer from MongoDB to PostgreSQL while preserving existing production data and ensuring a safe rollout.
- Worked closely with stakeholders to turn client-review feedback into concrete product improvements through rapid iteration.
- Built backend endpoints and business logic, resolved production issues, and contributed to architecture reviews covering service boundaries, communication patterns, migrations, and engineering trade-offs.

## Selected Project

### Tanilytics – Privacy-First Web Analytics Platform

2025 – 2026

TypeScript, React, Go, Spring Boot, Redpanda, ClickHouse, Redis, PostgreSQL, Kubernetes, Helm

- Co-Designed (team of 3) a privacy-first analytics platform ingesting 50K events/sec via Go/Redpanda, with GDPR-compliant IP anonymization (truncation → geolocation → SHA-256 hashing) and Redis Bloom filter deduplication.
- Built a polyglot microservices architecture (Go, Java/Spring Boot, TypeScript) across 7 services, achieving sub-500ms dashboard queries via ClickHouse materialized views and Redis real-time caching.
- Architected multi-tenant isolation and mitigated noisy-neighbor risk through per-tenant rate limiting (Kong), Redis-backed hot reads, and ClickHouse partitioning.

## Education

### INSAT – Institut National des Sciences Appliquées et de Technologie

Tunis, Tunisia

Engineering Degree in Software Engineering

2022 – 2027

Focus: software design, web engineering, distributed systems, databases, operating systems, algorithms, and applied machine learning.

Seeking a mandatory 6-month End-Of-Studies Internship starting February 2027.

## Technical Skills

**Languages:** TypeScript, JavaScript, Python, Java, Go

**Frontend:** React, Next.js, Tailwind CSS

**Backend:** FastAPI, Quarkus, Spring Boot

**Data & Messaging:** PostgreSQL, MongoDB, Redis, ClickHouse, Kafka, Redpanda, RabbitMQ

**Architecture:** Microservices, event-driven architecture, API design

**AI & DevOps:** LLM integration, RAG, Docker, Kubernetes, Helm, Github Actions

**Spoken Languages:** Arabic (Native), English (Full Professional), French (Full Professional), German (Elementary)

## Leadership & Volunteering

### INSAT ACM Student Chapter – Event Manager

2024 – 2025

Tunis, Tunisia

INSAT

- Coordinated technical events with **hundreds of participants**, managing logistics and volunteer teams.

### AIESEC Eskişehir, Türkiye – Exchange Participant

2024

Eskişehir, Türkiye

- Participated in **international programs** promoting Sustainable Development Goals through education.